St Joseph's College,
Gregory Terrace -
Human Movement Faculty

MOTOR SKILLS, MOVEMENT AND FEEDBACK

Year 8 Health and Physical Education

Term 4

Name: _____

Welcome to our PE unit on skill acquisition, movement, and feedback. Over the next few weeks, you'll discover how people learn and improve physical skills—from first attempts to expert performance. You'll explore the different stages of learning a skill and find out what helps athletes move more efficiently and effectively. You'll also see how feedback—from your teacher, your teammates, and even yourself—can help you improve and reach your sporting goals. By the end of this unit, you'll not only understand how to improve your own performance, but you'll also build the skills to support others through helpful and positive feedback.

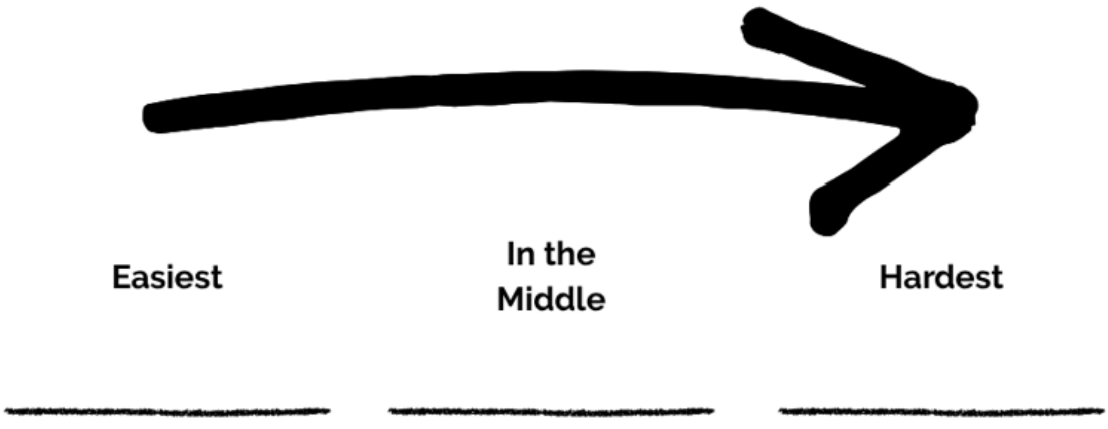
What is the current sporting unit you are participating in this Term in HPE?

_____

What are 3 skills from that sport?

1.
2.
3.

Rank these skills from easiest to perform to hardest to perform.



Class discussion.

"What makes a skill easier or harder to learn and perform successfully?"

Motor Skills

A **motor skill** is part of an activity that requires voluntary muscular movement to achieve a goal or task. Motor skills can be classified according to the following criteria (copy from Motor Skill PowerPoint):

1) _____
_____

2) _____
_____

3) _____
_____

Below, try to match the description of the classification to the appropriate criteria name:

DESCRIPTION	CRITERIA
1. How precise a movement is.	A. Open or Closed
2. Whether the environment affects the performance of the skill.	B. Discrete, Serial or Continuous
3. Whether the movement has a definite beginning and end.	C. Fine or Gross

Fine and Gross Motor Skills

Use information from the “Motor Skills” PowerPoint to complete the following information:

The gross – fine continuum distinguishes motor skills according to _____ _____.

Gross motor skills involve movements that require _____ muscle groups. They require _____ precision. Examples of gross motor skills include:

- _____
- _____
- _____

Fine motor skills involve _____movements that primarily require _____ muscle groups. The requires a _____ degree of precision. Examples of fine motor skills include:

- _____
- _____
- _____

Many skills involve both large and small muscles/muscle groups and cannot be defined exclusively as either fine or gross. That is why they are placed on a **continuum**. A continuum is something that changes gradually without clear steps or breaks. It’s like a smooth line that goes from one end to the other, with tiny changes along the way. So, a continuum is a way of showing how things can exist on a scale, not in separate boxes.

Categorise the following motor skills on the continuum below:

Throwing a dart Shot put Table tennis serve Back-flip AFL drop-punt



Open and Closed Motor Skills

Use information from the “Motor Skills” PowerPoint to complete the following information:

The closed – open continuum distinguishes motor skills based on the relative consistency or stability of the performance environment. The performance environment consists of:

- _____
- _____
- _____

A **closed skill** is performed in a highly _____, stationary/stable environment. Examples of a closed skill include:

- _____
- _____
- _____

An **open skill** is performed in a highly _____, moving/unstable environment. Examples of an open skill include:

- _____
- _____
- _____

Classify the following motor skills on the continuum below:

Driving a golf ball Surfing Throwing a dart AFL mark Bench press



Discrete, Serial & Continuous Motor Skills

The discrete-continuous continuum distinguishes motor skills on whether or not the movement has a specific beginning and ending.

A **discrete motor skill** has a clearly defined _____ and _____. Individually, come up with some examples of a discrete skill:

- _____
- _____
- _____

Continuous motor skills are typically _____ movements that don't have a clearly defined beginning or ending. Try to think of some examples of a continuous skill:

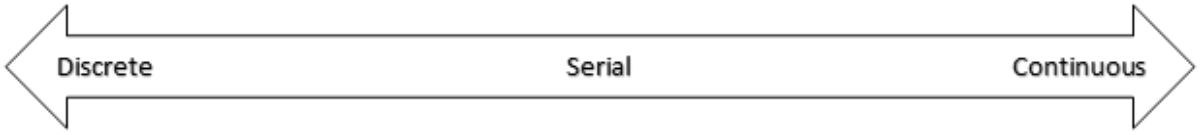
- _____
- _____
- _____

Discrete skills linked together to create a more complex continuous movement sequence are classified as **serial motor skills**. Discrete motor skills are typically _____ movements with specific _____ and _____ points. Examples include:

- _____
- _____
- _____

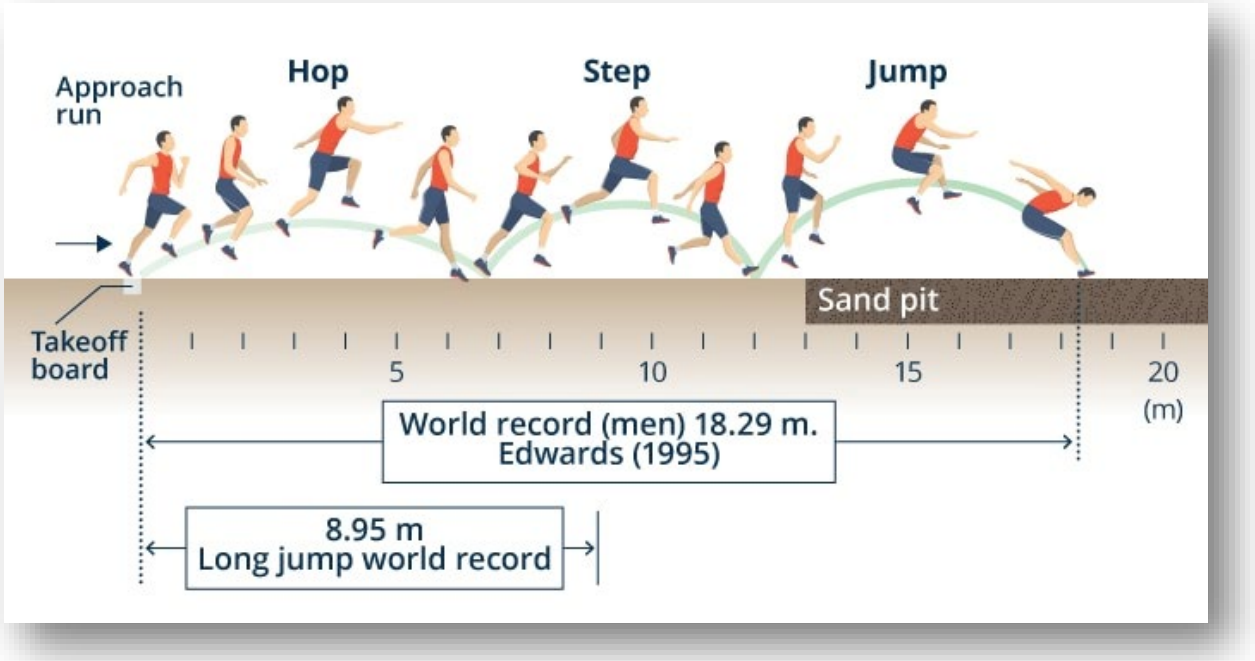
Classify the following skills on the continuum below:

Bowling a cricket ball Road cycling Basketball free throw Juggling



Use all three continua to describe each of the following skills. The first is done for you:

Rowing	Gross, closed and continuous
Badminton Serve	
Dribbling in futsal	
Triple jump	



Motor Skills, Movement Sequences, Movement Strategies, Principles of Play

In order to play a sport, we need to be able to do more than just complete basic motor skills. We need to be able to put motor skills together to create more complex series of movements, to achieve tasks or goals in the sport, in order to gain an advantage and ultimately put in our best effort to win.

Movement Sequences

Movement sequences are a series of motor skills that are combined to create a specialised movement for a particular sport, task or position. An example of a movement sequence would be a forehand in tennis.
Identify some important movement sequences from the four sports you have done in HPE this year.

Futsal	Basketball	AFL	Water Polo

Movement Strategies

A movement strategy is a unique movement sequence(s) that you select and perform to achieve a specific outcome or goal. An example of a movement strategy in tennis is performing a drop shot.
Identify some important movement strategies from the four sports you have done in HPE this year.

Futsal	Basketball	AFL	Water Polo

Principles of Play

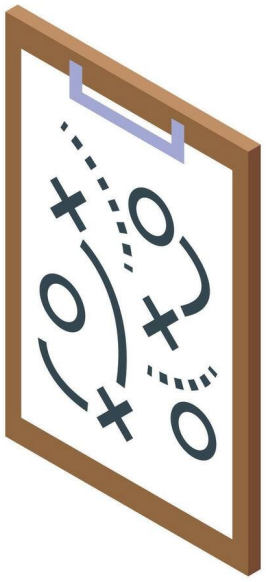
Principles of play are specific aspects of the game in which you must succeed to beat the opposition. Players apply a range of movement strategies to achieve success in principles of play. Principles of play are broken into the following categories:

- Defending
- Attacking
- Setting Up
- Using Space

Choose one sport that you have participated in this year in HPE and try to allocate movement strategies into the appropriate **principle of play**.

SPORT: _____

Defending	Attacking	Setting Up	Using Space



Skill Transfer in Sport

Playing more than one sport as a young athlete offers big benefits. Each sport teaches different movements, tactics and ways of thinking. These experiences build a broad base of physical, mental and social skills. Many of these abilities – such as balance, coordination, teamwork, decision-making and communication – can transfer from one activity to another. For example, footwork from basketball can help in netball, or hand-eye coordination from tennis can improve cricket. By trying different sports, athletes develop versatility, reduce overuse injuries and build confidence. This variety creates stronger, more adaptable players who are ready for future challenges.

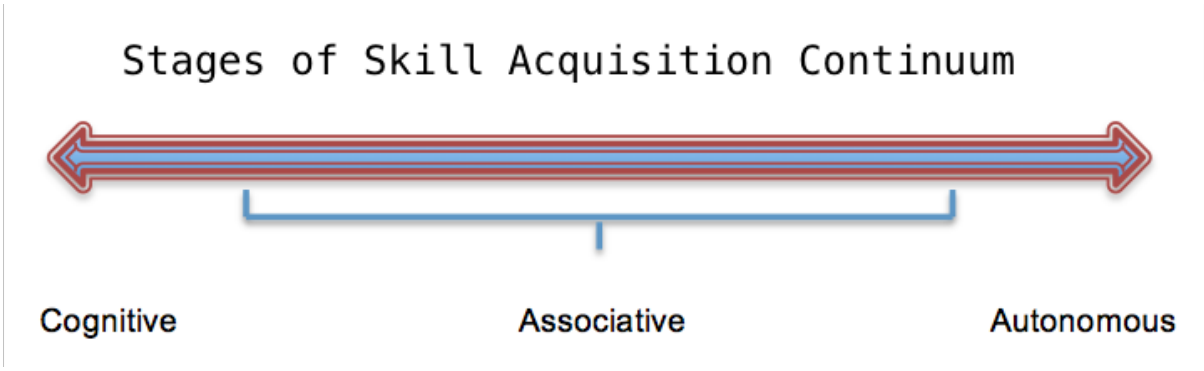
Activity

Instructions: Complete the table below by thinking about skills from different sports and how they positively transfer to help you experience success in another sport.

Sport 1	Skill	Sport 2 (where the skill is useful)	Why the skill transfers
Example: Volleyball	Blocking	Basketball	Developing vertical jump and timing will help improve shot blocking and rebounding

Stages of Learning

When we learn a new skill, we move through different stages as our ability improves. At first, in the **cognitive stage**, we are thinking hard about what to do and often make mistakes. With practice, we enter the **associative stage**, where movements become smoother and we start to correct errors ourselves. Finally, in the **autonomous stage**, the skill feels automatic, and we can focus on tactics or combining it with other skills. Understanding these stages helps athletes stay patient, practise effectively, and see how effort leads to progress over time.



Refer to the appropriate slides in the PowerPoint titled “Stages of Learning, Practice and Feedback” to complete the paragraphs below.

Cognitive Stage:

The cognitive stage of skill acquisition is characterised by _____ _____ and the athlete _____ about the skill. The athlete at this stage needs to think about their _____ _____, which _____ they are _____ and what the movement _____ _____ _____. The athlete is thinking about what they are doing at each _____ of skill execution, resulting in a _____ _____ movement.

Athletes at this stage have _____ _____ _____ and have a _____ jerky movement. Such an athlete may miss the ball that they are attempting to kick completely, or kick it backwards instead of forwards. Try and visualise a toddler learning to walk, or a 3 year old trying to kick a ball. They watch lots, try and mimic movements, and get frustrated at their errors.

Associative Stage:

The associative stage of skill acquisition is when the athlete has progressed from thinking about what they are _____ to thinking about _____ they do the skill. This means they are no longer thinking about _____ _____, but where they are passing the ball, or hitting the ball. They begin to think about _____ _____ rather than just on whether they manage to kick or hit the ball.

The associative stage of skill acquisition sees the movement become _____ _____ and _____. There are still _____ though as they move along the _____ these are not as _____ or as _____ as the cognitive stage of skill acquisition.

Autonomous Stage:

The autonomous stage of skill acquisition is when the athlete no longer thinks at all about the skill. In this stage the movement comes _____ and the athlete can _____ on other aspects of the competition, such as who to pass the ball to, how to beat the defensive player, or where to hit the forehand.

The autonomous stage of skill acquisition is the _____ stage and athletes who reaches this stage exhibit _____ of skilled performers. Such athlete's can _____ their own _____ midway through the movement to _____ to oppositional movements or _____ _____. They consistently perform the skill well with minor errors occurring rarely.



Practice

As the old saying goes, practice makes perfect. At least, it can get you closer to being perfect at a skill. Multiple studies have indicated that to reach elite levels in sport, arts and sciences, performers often accumulate more than 10 000 hours of practice. However, in real life most people do not have the time or motivation to complete such a large quantity of training. Therefore, practice needs to be of the highest quality possible in order to maximise its effect in improving performance.

There are a number of different types of practice to help an athlete improve their performance. The selection of the appropriate training type will depend on a number of variables including the skills being learnt, the level of experience of the learner and the performance environment. Following are a number of practice types. Fill in the blanks from the PowerPoint slides.

Massed Practice vs Distributed Practice

Massed practice involves practicing a skill with _____ or _____ rest for an _____ period.

Distributed practice involves practicing a skill in _____ with _____ periods in between.

Massed practice is suitable for _____ skilled performers with high levels of _____.

Distributed practice is suited to:

- Learners in the _____ stage
- Skills with low _____ demands
- _____ is low
- The task is _____

Part Practice vs Whole Practice

Part practice involves practicing a skill by breaking it down and repetitively practicing _____ or _____ of the skill.

Whole practice involves practicing a skill in its _____.

Practicing parts of a skill makes a complex task _____ for a _____. However, this is only effective if the parts of the skill are relatively _____ of each other (e.g. different elements of a gymnastics floor routine).

Whole practice is effective when the skill being learnt is relatively _____, but its parts are highly _____ on each other (e.g. serving in tennis).

Blocked Practice vs Random Practice

Blocked practice involves practicing _____ skill over and over again in a _____ environment for an _____ period of time.

Random practice involves practicing two or more skills in a _____ order.

Blocked practice leads to better performance of the motor skill within the _____ environment. Blocked practice may be useful in the early _____ stage of learning a skill to build familiarity with the skill and _____ in the learner.

Random practice leads to _____ lasting learning of the skill and better transfer of the skill to the _____ environment.



Reflect on your experiences with sport and HPE at Terrace. What types of practice have you largely been exposed to? Answer the following questions in the space below.

- What types of practice do you usually do when learning new skills or beginning a new practical unit?
- What type of practice do you participate in at the end of a HPE unit or in the middle of a sporting season?
- How would a training session differ from the 5D Volleyball Team to the 1st VI?

Complete the following table to the best of your ability:

Type of Practice	Learner Best Suited (cognitive, associative, autonomous) and Why
Massed	
Distributed	
Part	
Whole	
Blocked	
Random	

Feedback

Performance-related feedback is the _____
_____
_____

Intrinsic feedback is information you receive from your _____ when you move. It includes what you can _____, _____ and _____.
Proprioception is an awareness of _____ _____ through sensory receptors throughout the body.
Some examples of intrinsic feedback include:

- _____
- _____
- _____

Extrinsic feedback is information you receive about performance from an _____ source. There are two types of extrinsic feedback, knowledge of _____ and knowledge of _____.
Knowledge of results is feedback about the degree of _____ of your movements.
Knowledge of performance is feedback about the _____ or technical _____ of your movements. This often comes from a coach or teacher. _____ replays are also useful sources of knowledge of performance feedback.

It is important to take into consideration the level of ability, personality and level of motivation of the learner when deciding what types and what amount of feedback to give them.

What type of feedback would you give a beginner tennis player who struggles to land their serves in? _____
_____

What type of feedback would you give an elite sprinter whose body position is a little bit too upright? _____
_____

EXAM PREPARATION

Feedback and Practice at Different Stages of Learning

Cognitive Stage:

A coach needs to provide lots of feedback and demonstrations during the cognitive stage of skill acquisition. They might use videos or other visualisations to help show the athlete what the skill looks like when done well. A coach will often break the skill down into its various sections to be put together as the athlete progresses in their learning.

As is often the case in learning a new skill, frequent short periods of exposure are best for its development. 20-40 min 3-5 times a week is a great amount of practice for learning a new skill. Even the training session itself, could break this new skill up by including skills the athlete does well, to ensure they are getting positive results as well as they learn the new skill. This makes distributed part practice the best practice method for the athlete at the cognitive stage of skill acquisition.

Describe a training drill or session that would benefit an athlete at the cognitive stage of a sport of your choice:

Associative Stage:

During this stage the athlete can begin to provide some of their own feedback, but they still benefit from immediate feedback concerning their technique provided by a coach as well as knowledge of results. The athlete can adjust their technique and begin to increase the complexity of the context in which the skill is executed. For example, hitting or kicking a moving ball, rather than a stationary one.

During the associative stage the athlete needs lots of practice that is whole and normally massed, though if they get bored distributed should be used. This stage often lasts a long time, with many athletes not progressing to the final autonomous stage of skill acquisition.

Describe a training drill or session that would benefit an athlete at the associative stage of a sport of your choice:

Autonomous Stage:

The autonomous athlete knows what the movement feels like and can provide their own feedback, though external feedback on skill execution may be beneficial. Coaching an autonomous athlete usually focuses on the execution of the skill under pressure and with various cognitive processes being completed at the same time. This is usually done through small-sided games or competition simulations, such as sparing.

Describe a training drill or session that would benefit an athlete at the autonomous stage of a sport of your choice: